InternSync — Diagram pack

Requirements and design diagrams · consistent with SRS v1.0 and the SPMP

No.	Diagram	What it shows	Where to use it
1	Use case diagram (p. 2)	Actors, use cases UC-01 to UC-15, include / extend relations	SRS 3.3 (above Table 3.1)
2	Context diagram (p. 3)	System boundary, users and external services	SRS 3.1 (Figure 3.1)
3a	Activity diagram — setup and placement (p. 4)	Swimlanes from opening a cycle to a confirmed placement	SRS 3.2 or design document
3b	Activity diagram — training and grading (p. 5)	Weekly loop, final evaluation, grade approval	SRS 3.2 or design document
4a	State machines — report and grade (p. 6)	Report slot and grade life cycles	SRS 4.9 (Figures 4.1, 4.2)
4b	State machines — vacancy and application (p. 7)	Vacancy and application / placement life cycles	SRS 4.4
5	System architecture (p. 8)	Layers, API modules, AI services, scheduler, data	Design document (T2), presentation
6a	ERD — accounts, cycles, vacancies, matching (p. 9)	Entities, keys and cardinalities (crow's foot)	Design document (T2)
6b	ERD — training, evaluation, grading (p. 10)	Reports, attendance, evaluations, grades, certificates	Design document (T2)
7a	Sequence — CV upload and matching (p. 11)	Skill extraction, confirmation, ranking with fallback	Design document
7b	Sequence — weekly report and review (p. 12)	Submission, Late flag, similarity check, accept / revision	Design document
7c	Sequence — grade submission and approval (p. 13)	Preconditions, approve / return, notifications	Design document
7d	Sequence — scheduled reminders and escalation (p. 14)	15-minute jobs, idempotent notifications	Design document
8	AI pipelines (p. 15)	Fit-score formula, explanation, similarity method (proposed)	Design document, presentation
9	Deployment options (p. 16)	Managed services vs a single VM	Design document (deployment), team decision
10	Schedule (Gantt) (p. 17)	Activities T1–T9, milestones M1–M7, dependencies	SPMP, progress presentations
11	Requirements by area and priority (p. 18)	116 requirements by area: Must / Should / Could	SRS 9.2

Colour key used throughout: blue = system behaviour or people; orange = AI-supported (advisory) behaviour; green = an approved / final state or the recommended option; grey dashed = external service.
Diagram 8 contains proposed design values (fit-score weights, similarity thresholds) — they are configurable and should be confirmed on the labelled test sets. Diagram 1 adds UC-15 (scheduler), which SRS Table 3.1 does not list yet. Editable sources (Python, Graphviz, Mermaid) and SVG files are in the accompanying zip.

Figure 1 — Use case diagram

Actors, use cases UC-01 to UC-15, include / extend relations. Use in: SRS 3.3 (above Table 3.1).

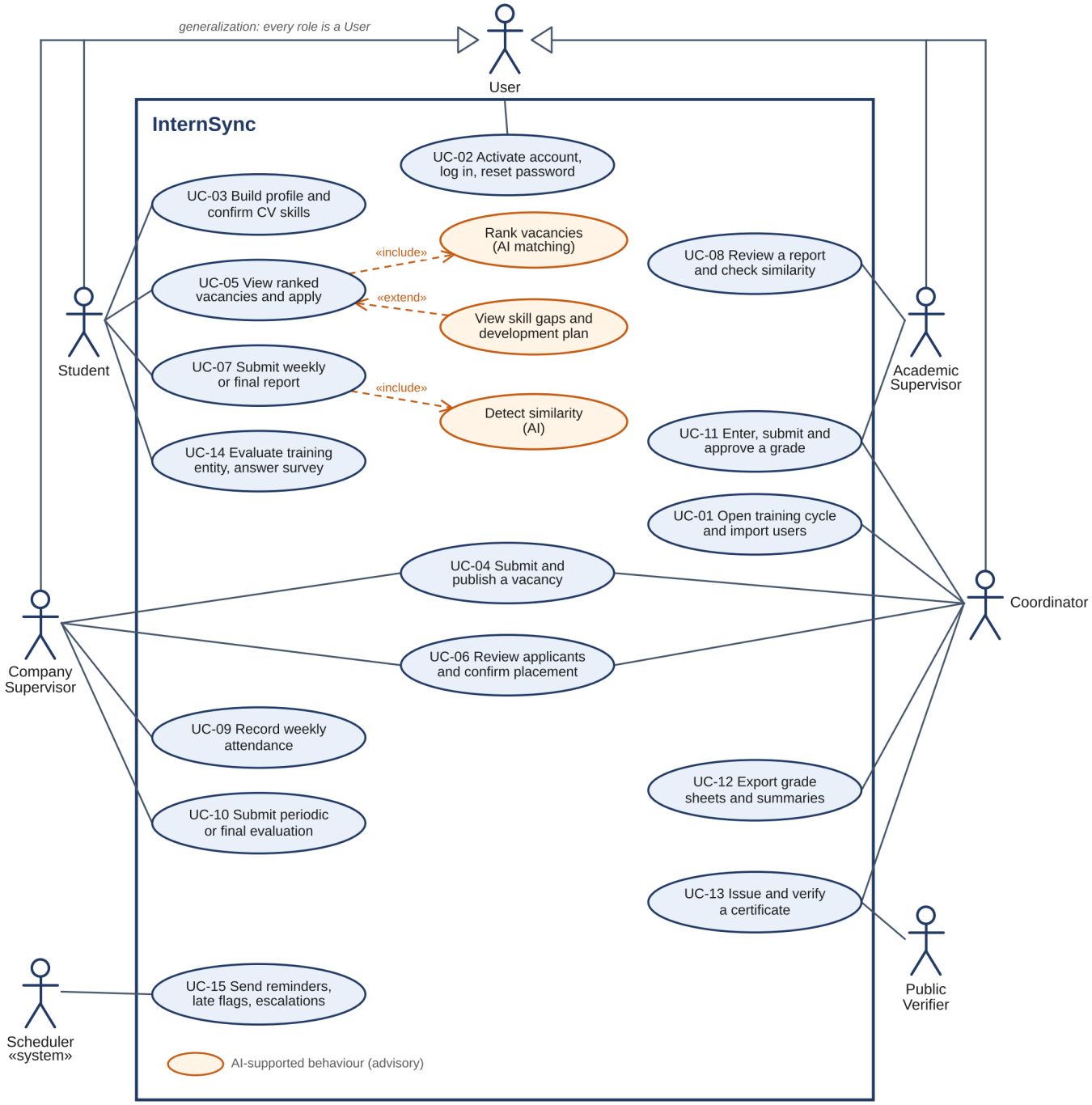


Figure 2 — Context diagram

System boundary, users and external services. Use in: SRS 3.1 (Figure 3.1).

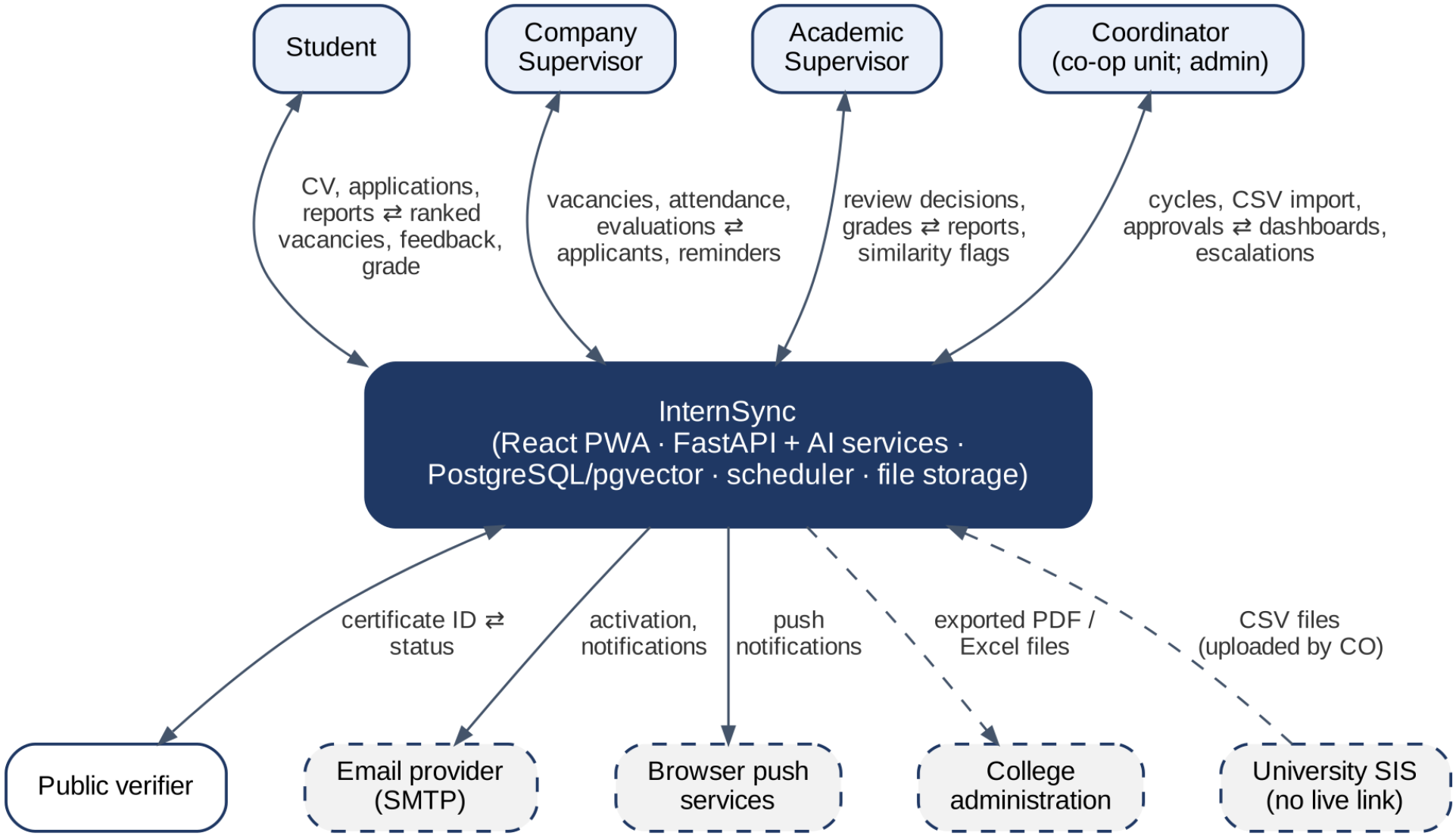


Figure 3a — Activity diagram — setup and placement

Swimlanes from opening a cycle to a confirmed placement. Use in: SRS 3.2 or design document.

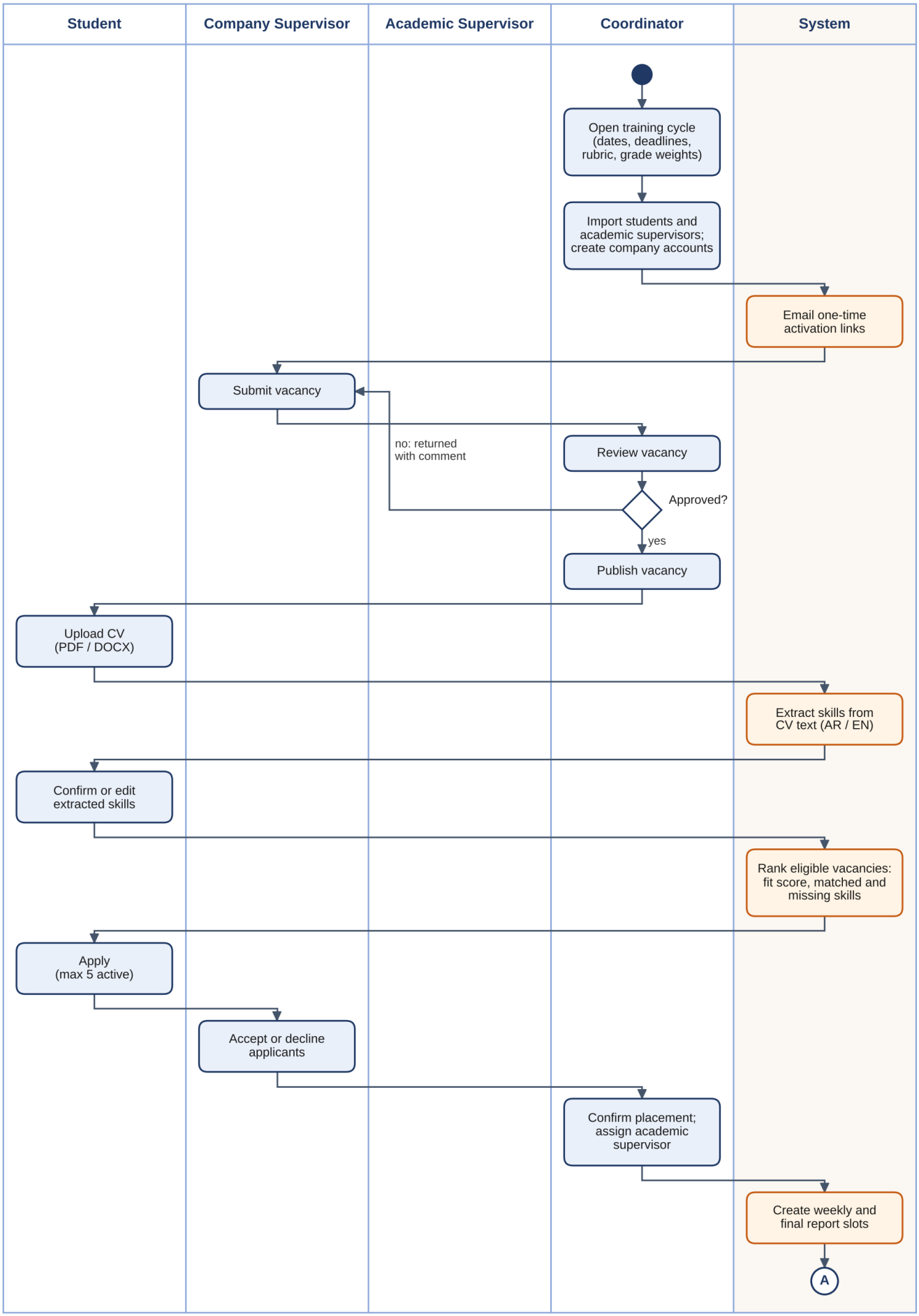


Figure 3b — Activity diagram — training and grading

Weekly loop, final evaluation, grade approval. Use in: SRS 3.2 or design document.

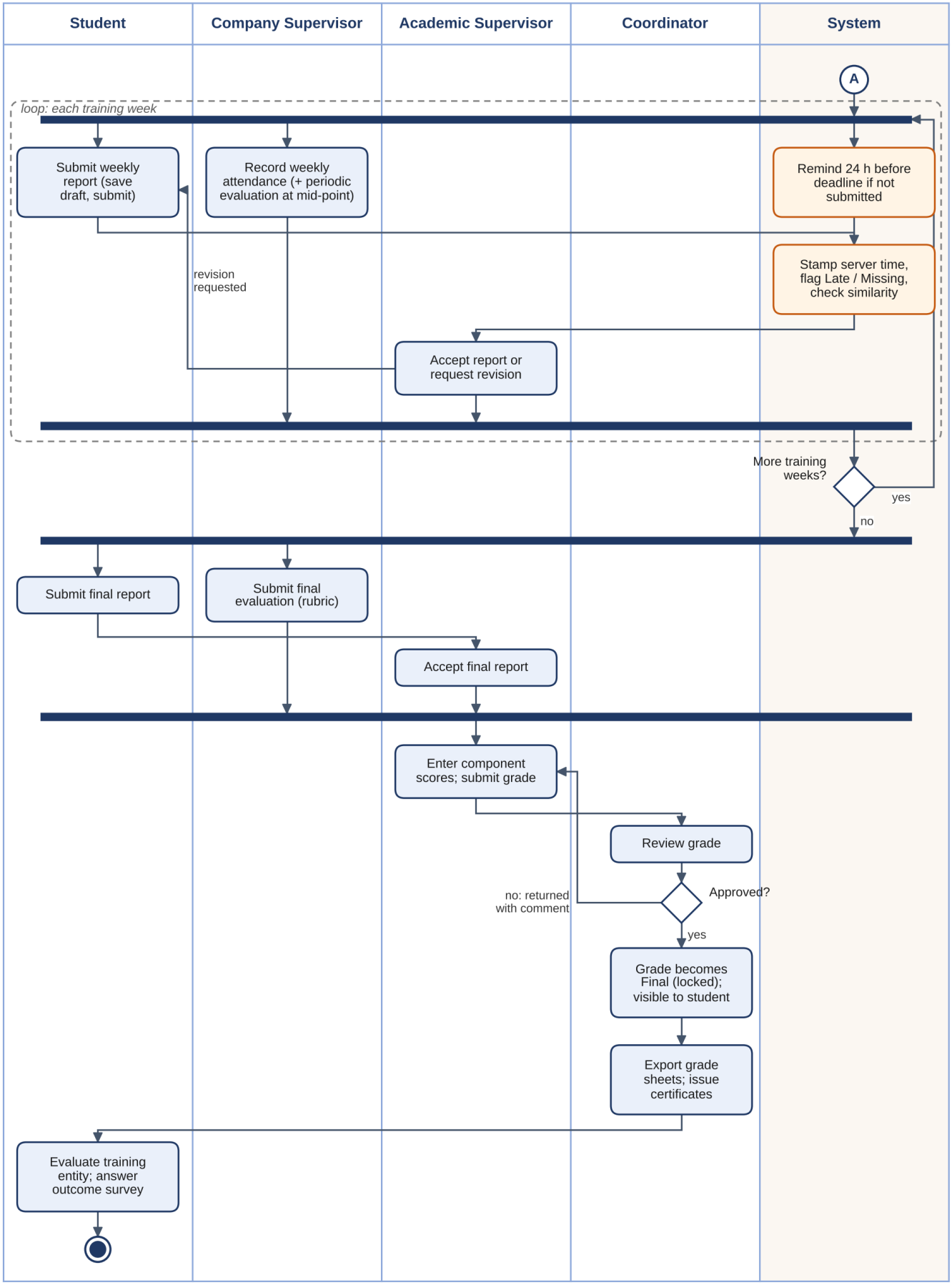
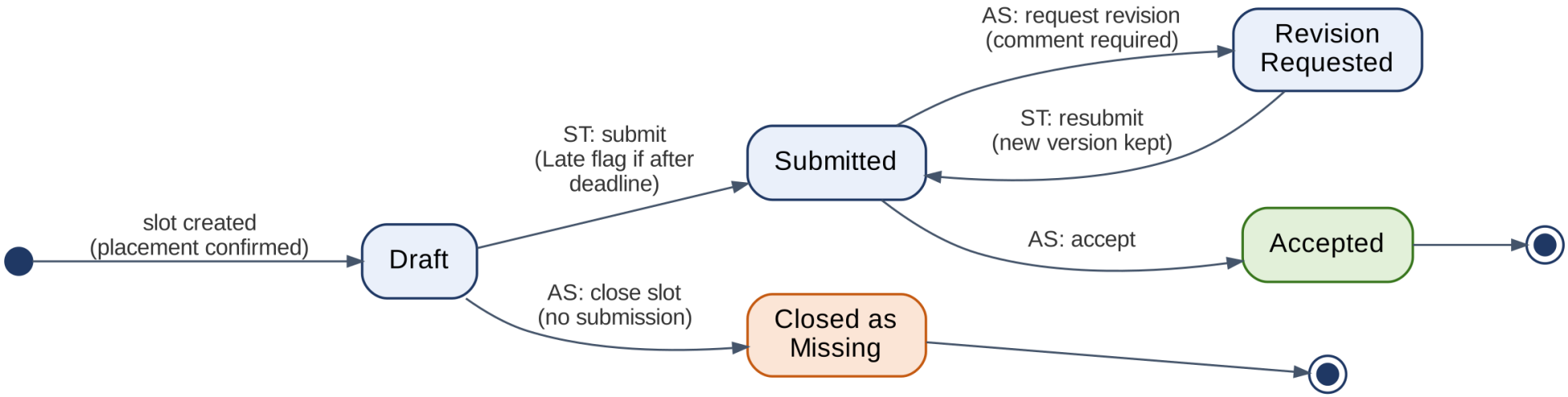


Figure 4a — State machines — report and grade

Report slot and grade life cycles. Use in: SRS 4.9 (Figures 4.1, 4.2).

Report slot (FR-WFL-01)



Grade (FR-WFL-03 – 05)

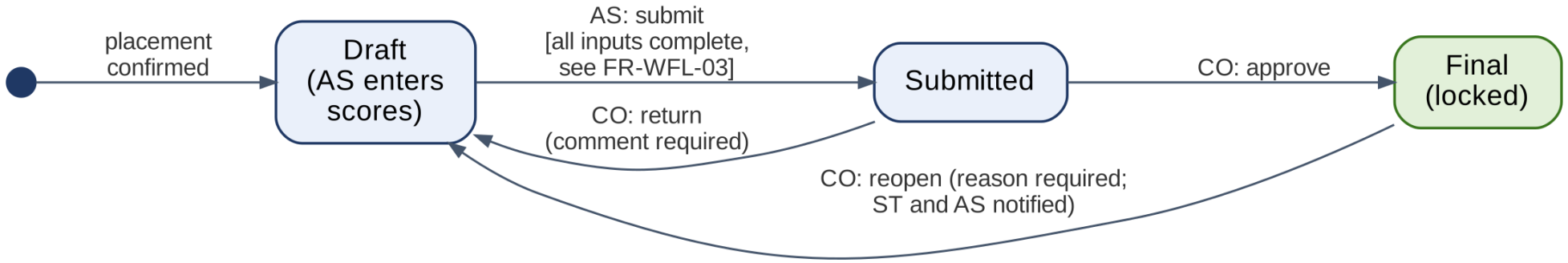
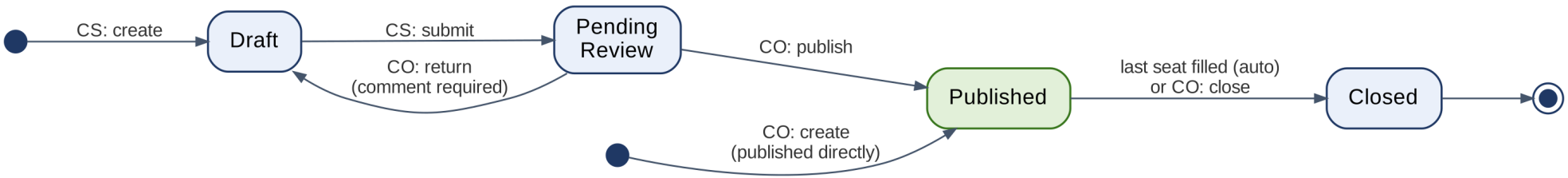


Figure 4b — State machines — vacancy and application

Vacancy and application / placement life cycles. Use in: SRS 4.4.

Vacancy (FR-VAC-02, FR-VAC-07)



Application and placement (FR-VAC-04 – 08)

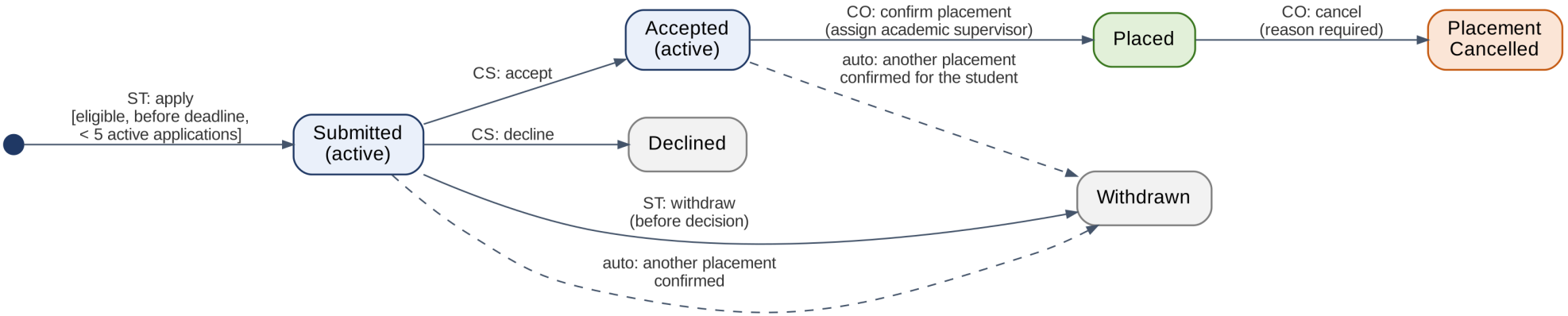


Figure 5 — System architecture

Layers, API modules, AI services, scheduler, data. Use in: Design document (T2), presentation.

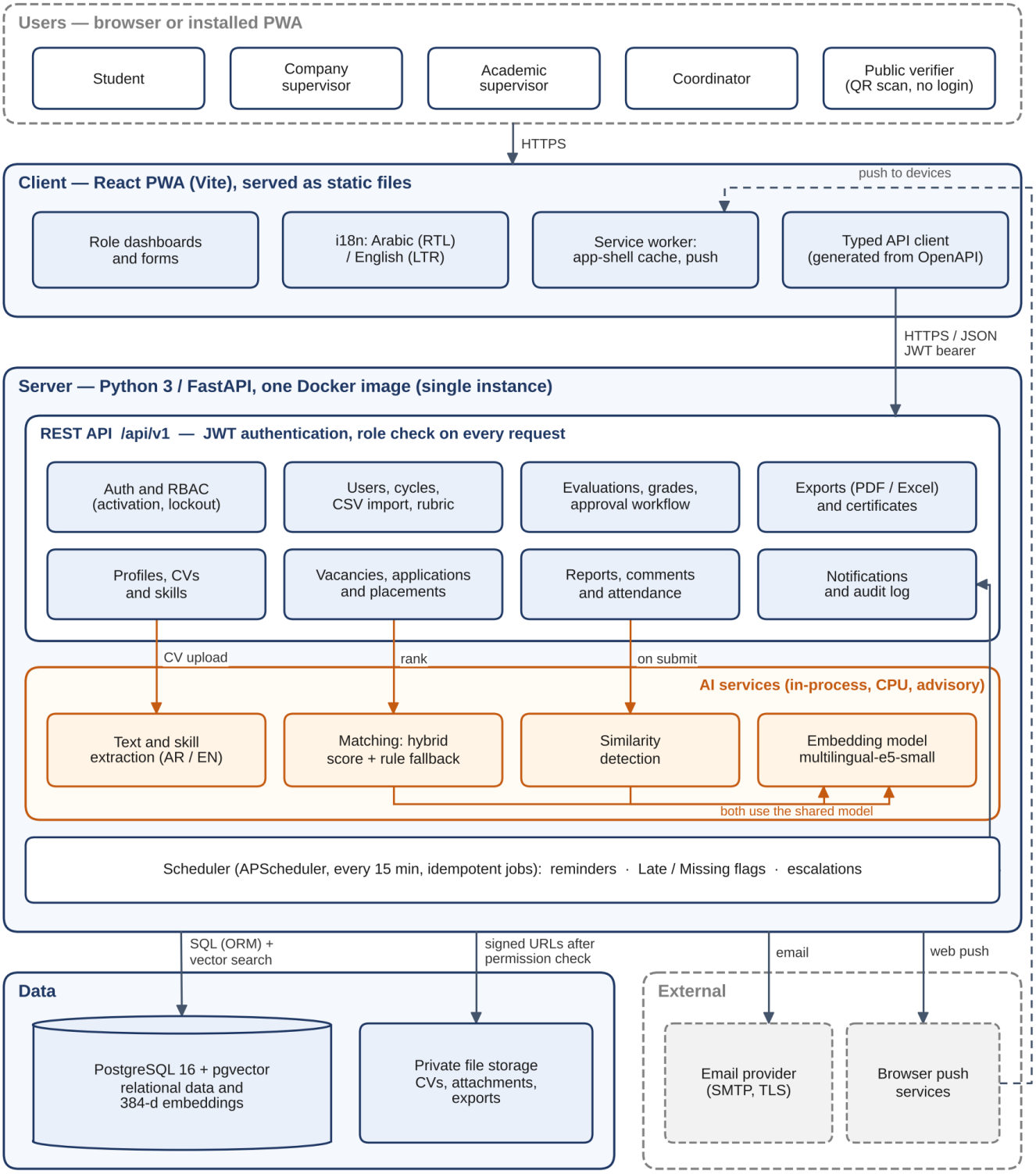


Figure 6a — ERD — accounts, cycles, vacancies, matching

Entities, keys and cardinalities (crow's foot). Use in: Design document (T2).

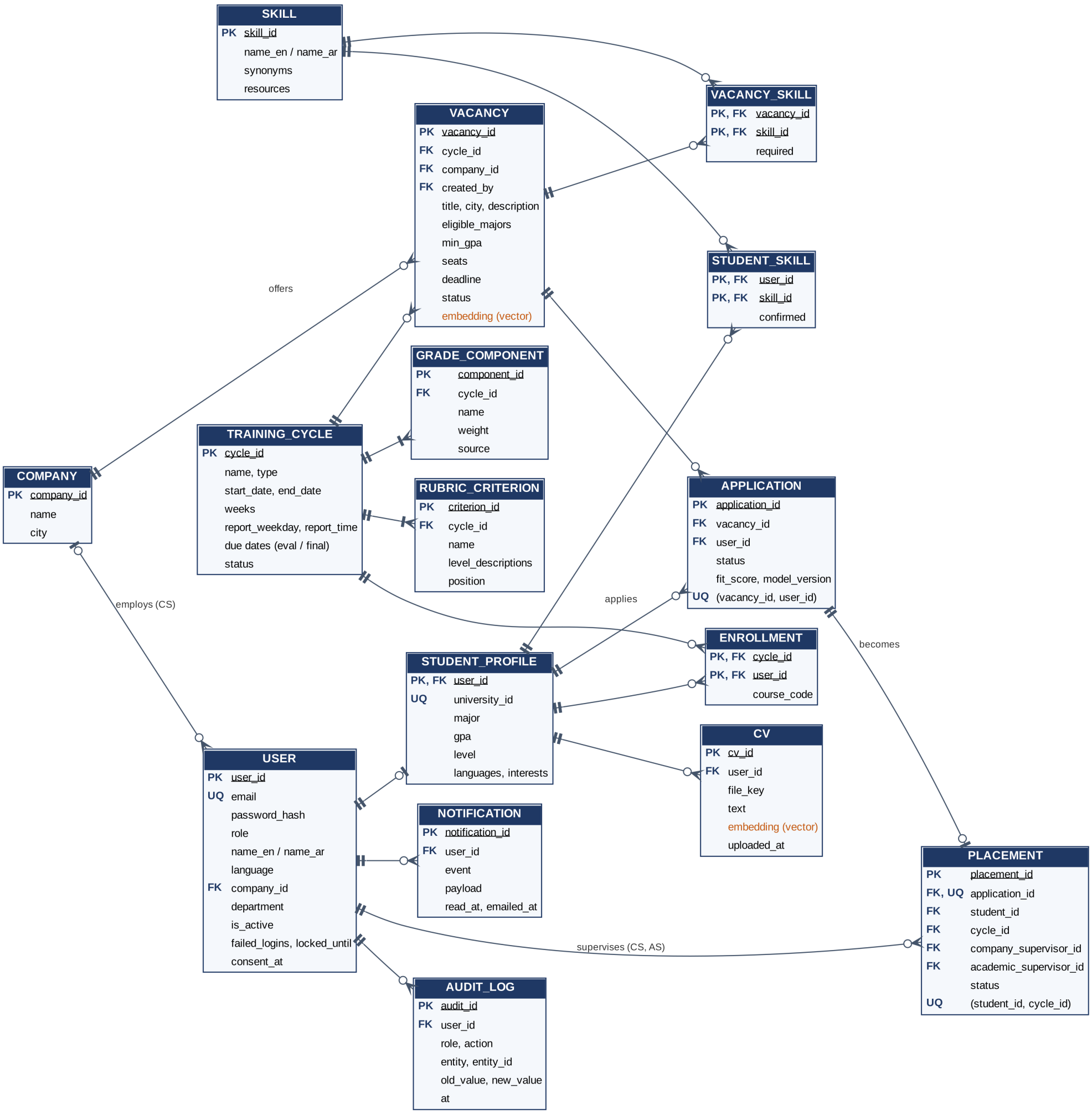


Figure 6b — ERD — training, evaluation, grading

Reports, attendance, evaluations, grades, certificates. Use in: Design document (T2).

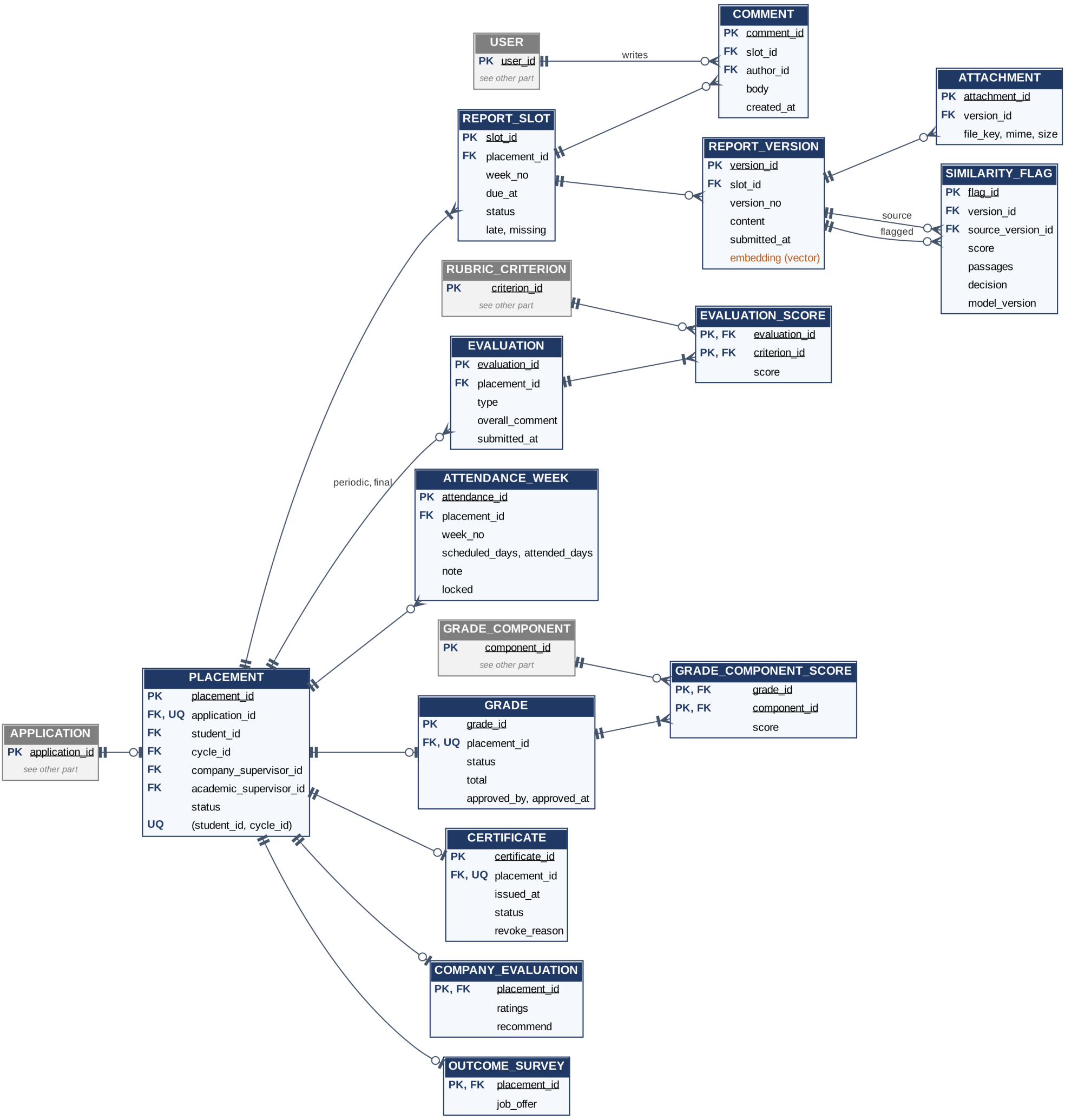


Figure 7a — Sequence — CV upload and matching

Skill extraction, confirmation, ranking with fallback. Use in: Design document.

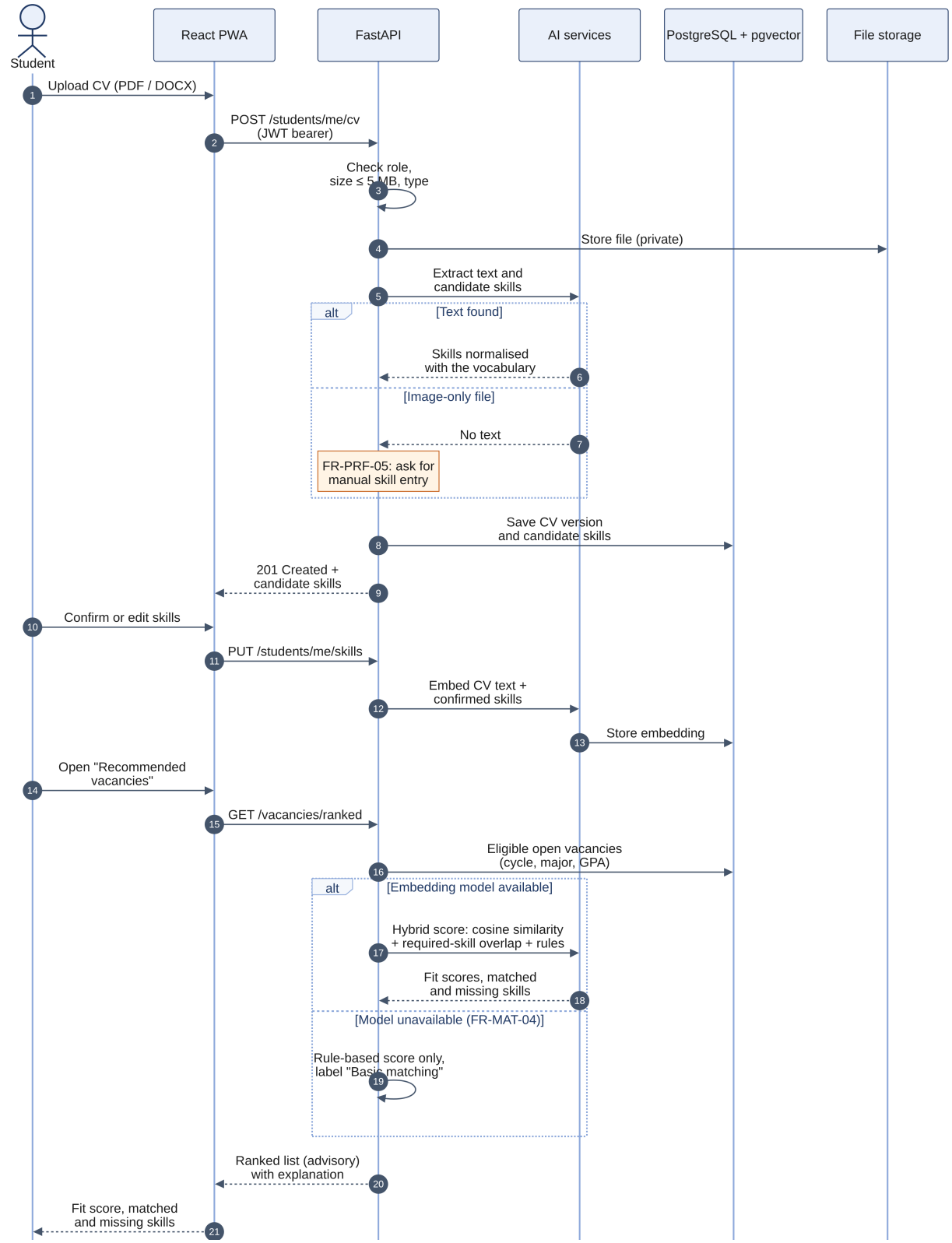


Figure 7b — Sequence — weekly report and review

Submission, Late flag, similarity check, accept / revision. Use in: Design document.

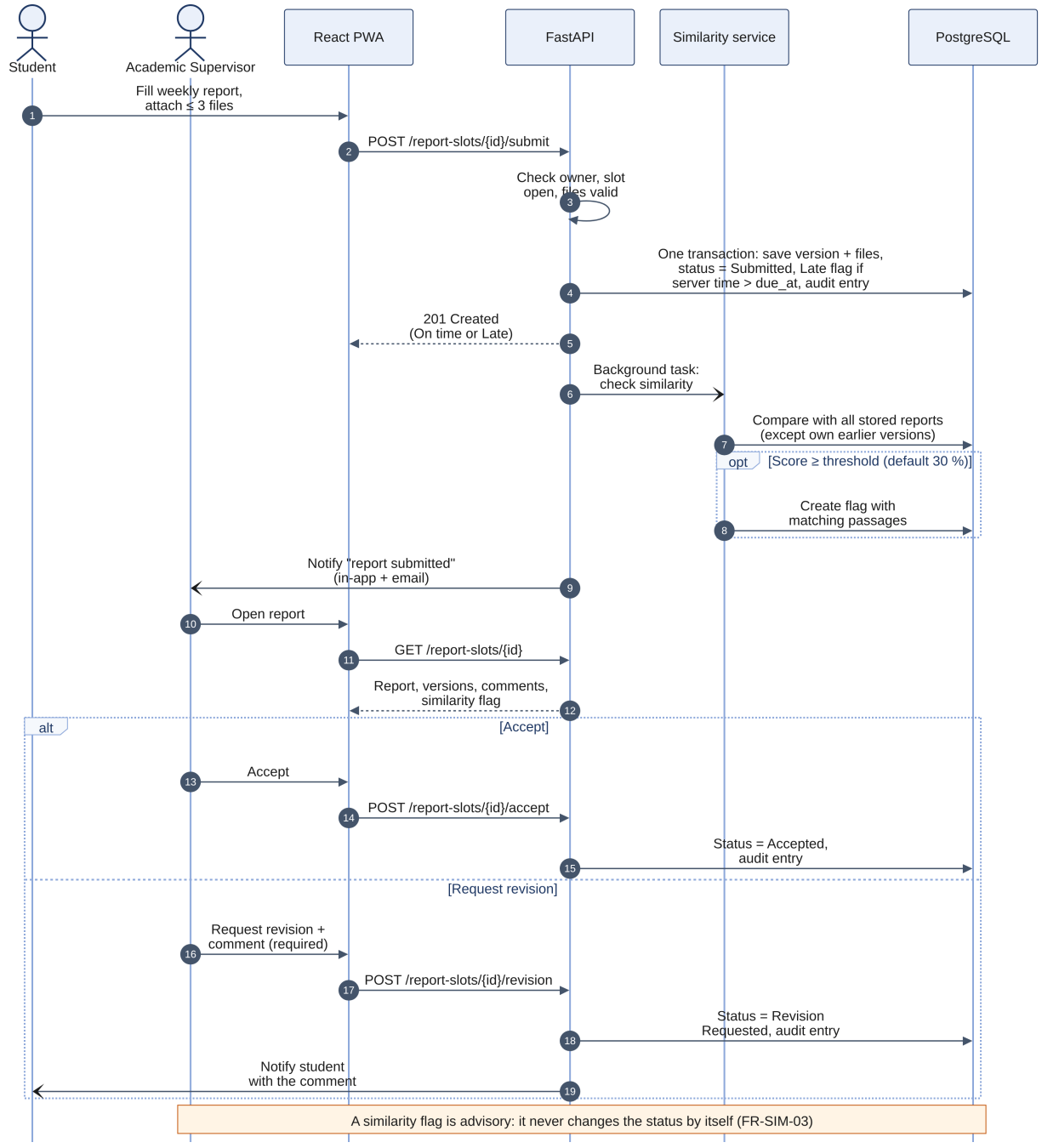


Figure 7c — Sequence — grade submission and approval

Preconditions, approve / return, notifications. Use in: Design document.

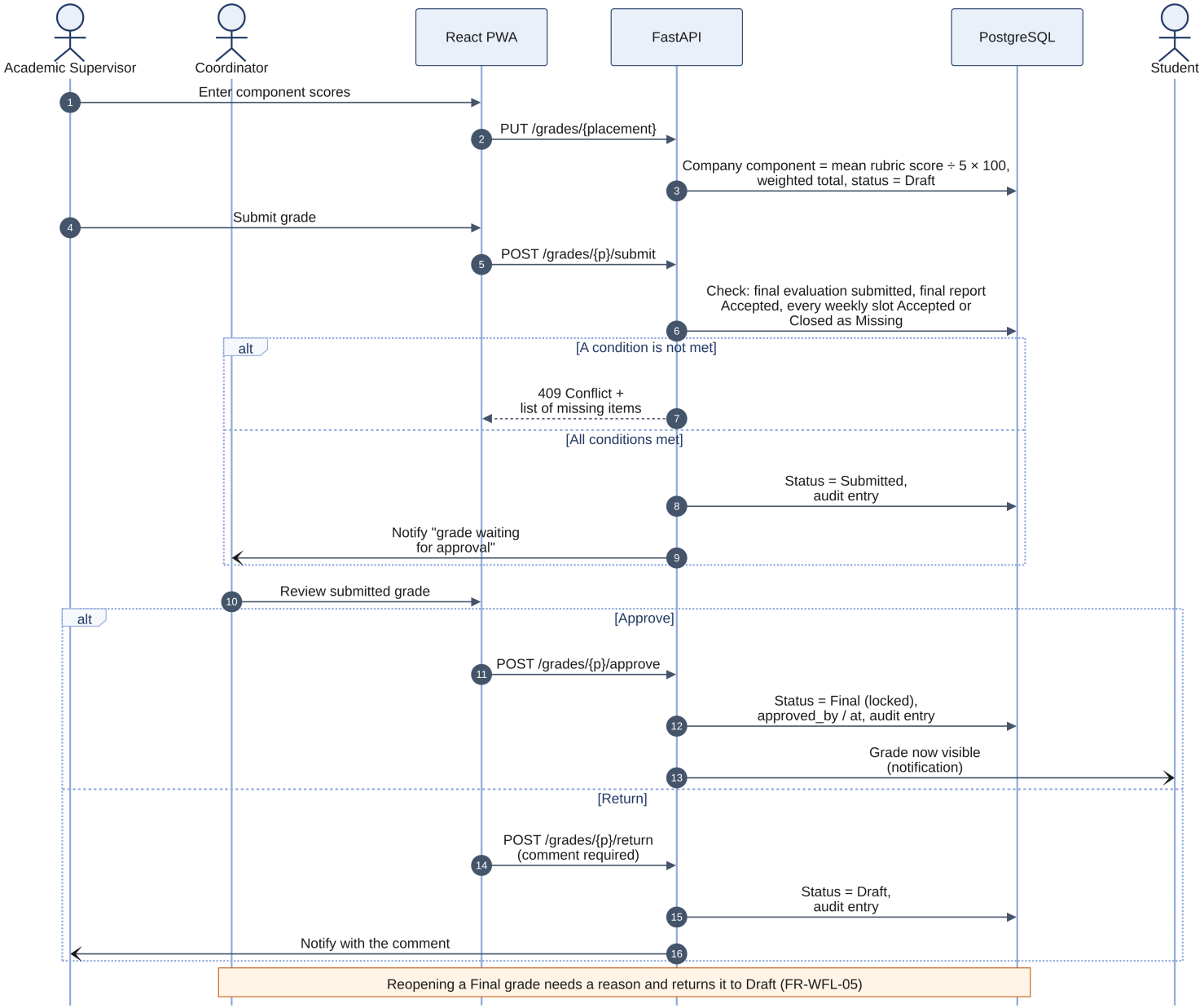


Figure 7d — Sequence — scheduled reminders and escalation

15-minute jobs, idempotent notifications. Use in: Design document.

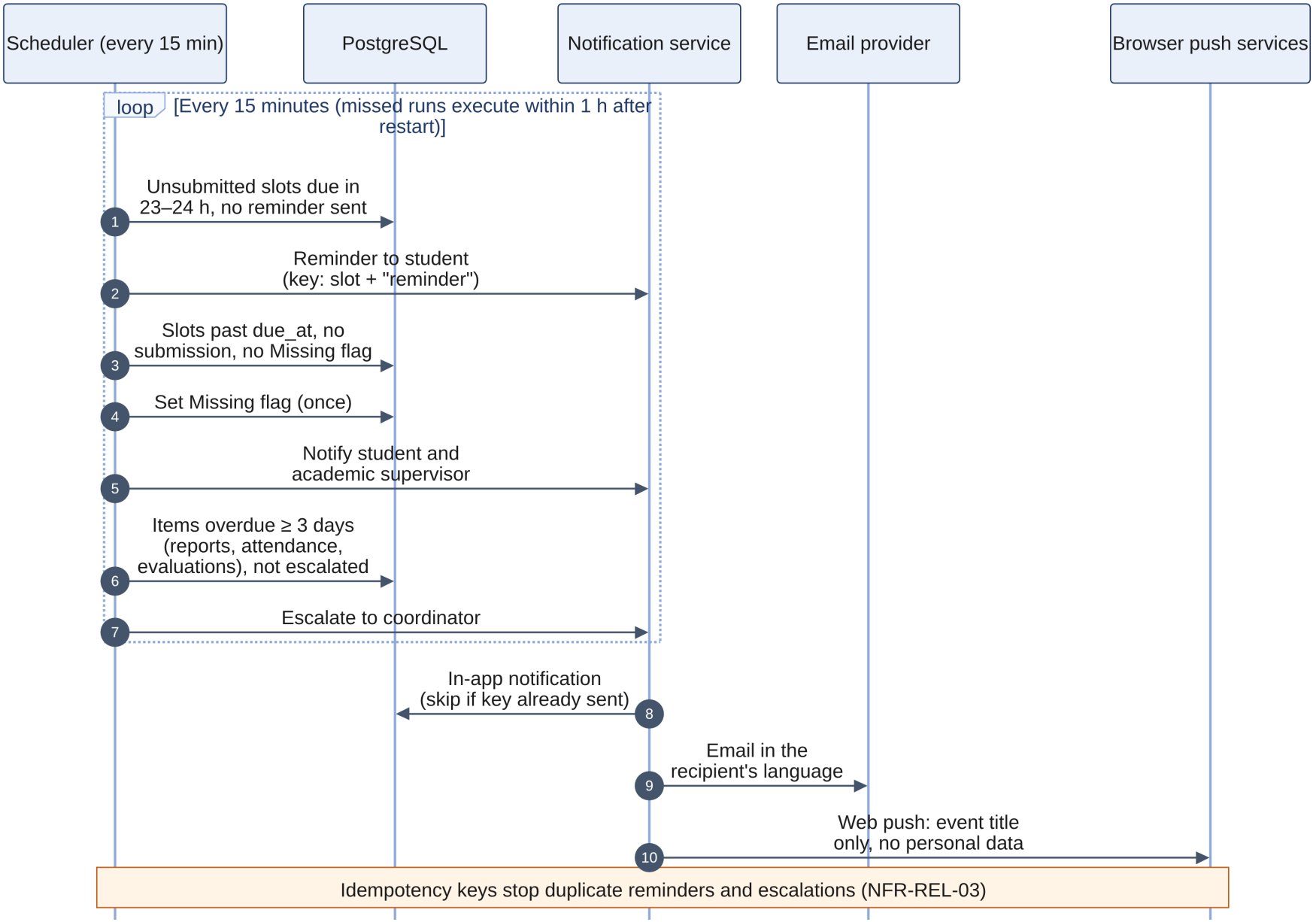


Figure 8 — AI pipelines

Fit-score formula, explanation, similarity method (proposed). Use in: Design document, presentation.

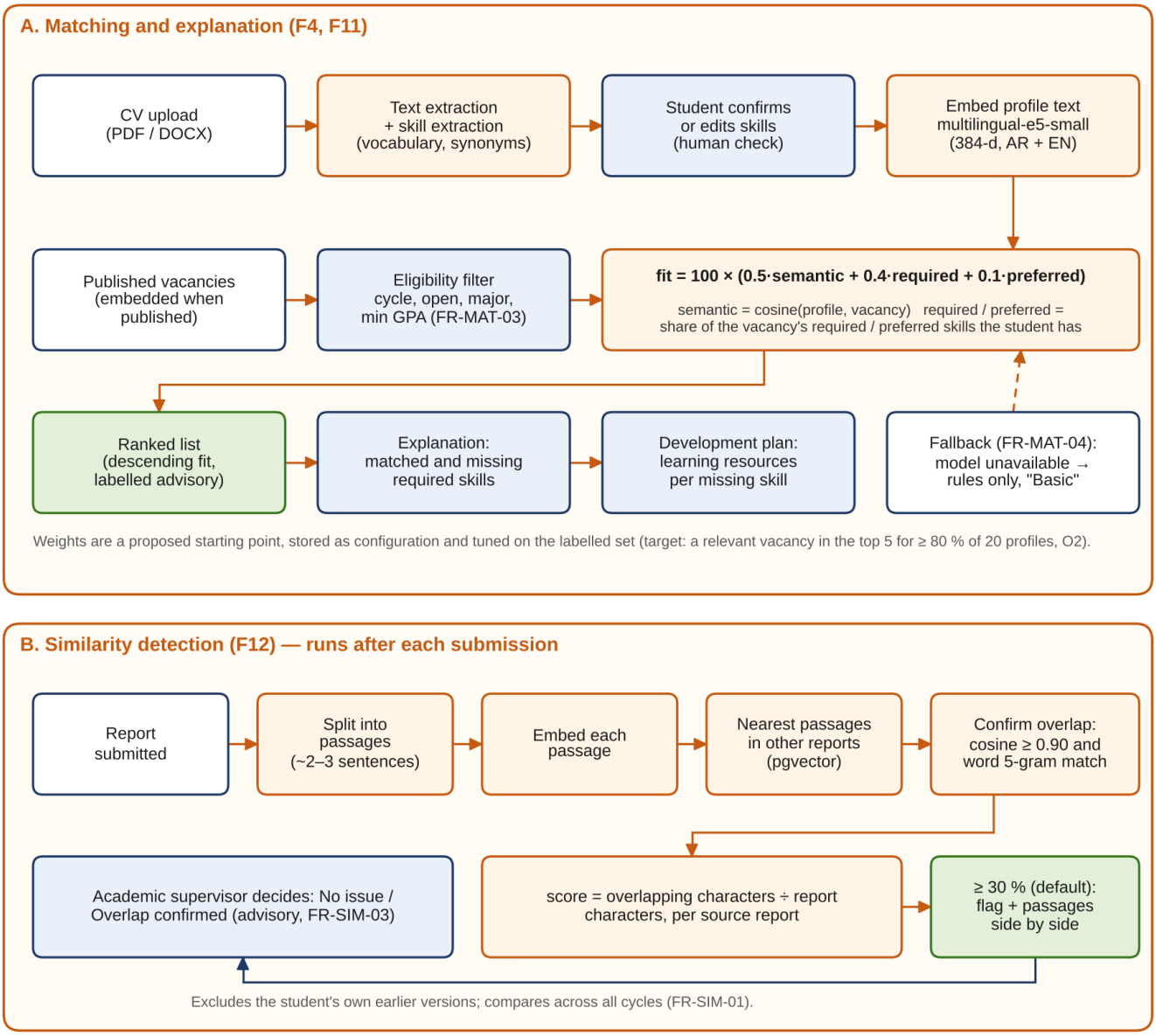


Figure 9 — Deployment options

Managed services vs a single VM. Use in: Design document (deployment), team decision.

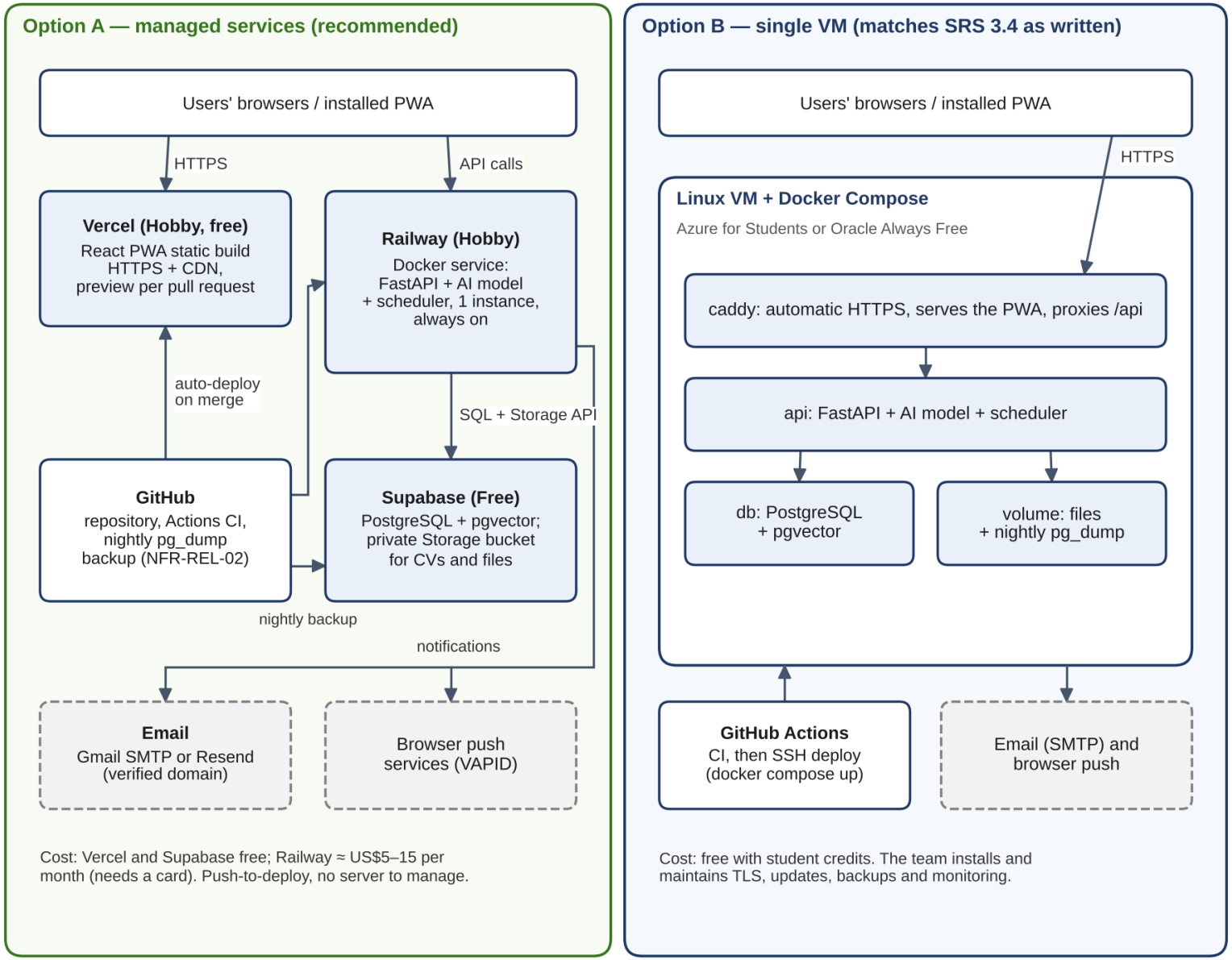


Figure 10 — Schedule (Gantt)

Activities T1–T9, milestones M1–M7, dependencies. Use in: SPMP, progress presentations.

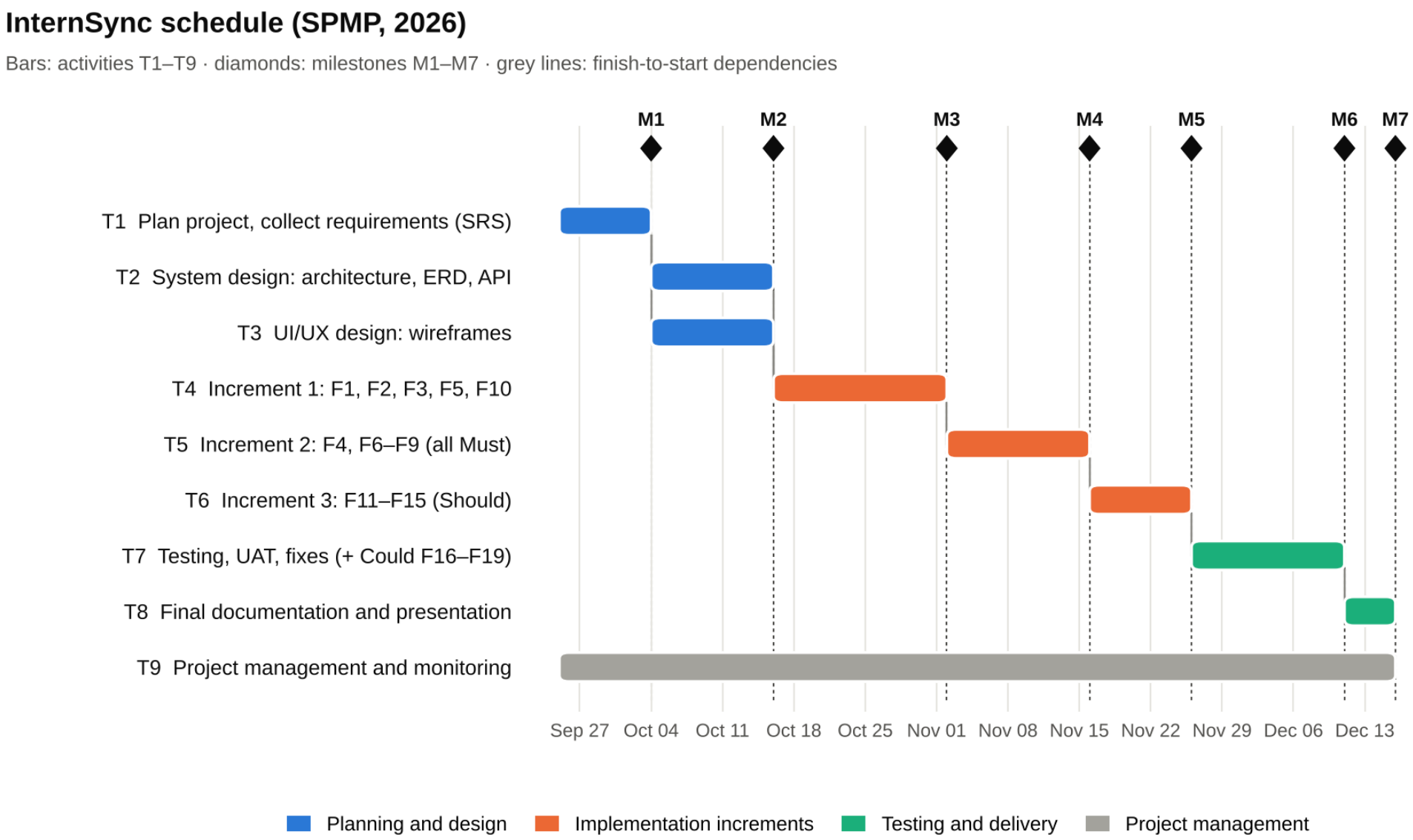


Figure 11 — Requirements by area and priority

116 requirements by area: Must / Should / Could. Use in: SRS 9.2.

